WORKED SOLUTION

STEP I 1997, Question 6

Pure • Integral Bounds for Pi

STEP I 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.3, *22/7 Is Bigger Than* π: https://beyondstep.keithmcnulty.org/ch04/twenty-two-sevenths.html

The key idea

The integrand is a rational function whose numerator has higher degree than its denominator, so the first job is polynomial division by x^2+1: a polynomial quotient, which integrates trivially, plus a constant remainder over $1+x^2$, which integrates to an arctangent. The arctangent is where π comes from. For the final inequalities, compare the integrand with $x^4(1-x)^4$ itself, using $1 \le 1+x^2$ on $[0,1]$.

Solution

The division

Expanding by the binomial theorem,

$$x^4(1-x)^4 = x^4(1-4x+6x^2-4x^3+x^4) = x^8-4x^7+6x^6-4x^5+x^4.$$

We divide by x^2+1, peeling off the leading term each time:

$$\begin{aligned} x^8-4x^7+6x^6-4x^5+x^4 &= x^6(x^2+1)+(-4x^7+5x^6-4x^5+x^4)\\ -4x^7+5x^6-4x^5+x^4 &= -4x^5(x^2+1)+(5x^6+x^4)\\ 5x^6+x^4 &= 5x^4(x^2+1)+(-4x^4)\\ -4x^4 &= -4x^2(x^2+1)+4x^2\\ 4x^2 &= 4(x^2+1)-4. \end{aligned}$$

Collecting the multiples of x^2+1,

$$x^4(1-x)^4 = (x^6-4x^5+5x^4-4x^2+4)(x^2+1)-4,$$

so

$$a_6 = 1, \quad a_5 = -4, \quad a_4 = 5, \quad a_3 = 0, \quad a_2 = -4, \quad a_1 = 0, \quad a_0 = 4, \quad b = -4.$$

A quick check: at $x = 0$ the right-hand side is $4 \cdot 1 - 4 = 0$, and at $x = 1$ it is $(1 - 4 + 5 - 4 + 4) \cdot 2 - 4 = 0$, matching the left-hand side both times.

The integral equal to $\frac{22}{7} - \pi$

Dividing the identity by $1 + x^2$ (which is never zero),

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Integrating term by term over $[0, 1]$, and using $\int \frac{1}{1+x^2}\,\mathrm{d}x = \arctan x$,

$$\begin{aligned}\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x &= \left[\tfrac{1}{7}x^7 - \tfrac{2}{3}x^6 + x^5 - \tfrac{4}{3}x^3 + 4x\right]_0^1 - 4\Big[\arctan x\Big]_0^1 \\ &= \tfrac{1}{7} - \tfrac{2}{3} + 1 - \tfrac{4}{3} + 4 \;-\; 4 \cdot \tfrac{\pi}{4} \\ &= \tfrac{1}{7} + 3 - \pi \\ &= \frac{22}{7} - \pi,\end{aligned}$$

since $-\frac{2}{3} - \frac{4}{3} = -2$ and $\arctan 1 = \frac{\pi}{4}$. This is the required result.

The integral of $x^4(1-x)^4$

Using the expansion from the start,

$$\int_0^1 x^4(1-x)^4\,\mathrm{d}x = \tfrac{1}{5} - \tfrac{4}{6} + \tfrac{6}{7} - \tfrac{4}{8} + \tfrac{1}{9}.$$

Over the common denominator $2520 = 5 \cdot 7 \cdot 8 \cdot 9$,

$$\frac{504 - 1680 + 2160 - 1260 + 280}{2520} = \frac{4}{2520} = \frac{1}{630}.$$

The bounds on π

Write $f(x) = x^4(1-x)^4$, so $f(x) > 0$ for $0 < x < 1$.

Upper bound. On $(0, 1)$ the integrand $f(x)/(1+x^2)$ is strictly positive and continuous, so its integral over $[0, 1]$ is strictly positive. Hence $\frac{22}{7} - \pi > 0$, that is, $\pi < \frac{22}{7}$.

Lower bound. For $0 < x < 1$ we have $1 + x^2 > 1$, and since $f(x) > 0$,

$$\frac{f(x)}{1+x^2} < f(x).$$

Ask yourself Why is the inequality between the integrals strict, not just $\leq$?

The difference $f(x) - \dfrac{f(x)}{1+x^2} = \dfrac{x^2 f(x)}{1+x^2}$ is continuous and strictly positive on the whole open interval $(0,1)$, so its integral over $[0,1]$ is strictly positive. (A continuous function that is non-negative and positive somewhere has positive integral.) Therefore

$$\frac{22}{7} - \pi = \int_0^1 \frac{f(x)}{1+x^2}\,\mathrm{d}x < \int_0^1 f(x)\,\mathrm{d}x = \frac{1}{630},$$

which rearranges to $\pi > \frac{22}{7} - \frac{1}{630}$. Combining the two bounds,

$$\frac{22}{7} > \pi > \frac{22}{7} - \frac{1}{630},$$

as required.

Checks

- Numerically, $\frac{22}{7} - \pi \approx 0.0012645$, and Simpson's rule on the first integral gives the same value to twelve decimal places.
- $\frac{1}{630} \approx 0.0015873$, comfortably bigger than 0.0012645; the bounds say $3.141270 < \pi < 3.142857$, and indeed $\pi \approx 3.141593$.
- The value $\frac{1}{630}$ also follows from the beta-integral formula $\int_0^1 x^m(1-x)^n\,\mathrm{d}x = \dfrac{m!\,n!}{(m+n+1)!}$, which gives $\dfrac{4!\,4!}{9!} = \dfrac{576}{362880} = \dfrac{1}{630}$.

Common slips

- Losing the two zero coefficients a_3 and a_1 in the long division: keep every power in place, or the remainder comes out wrong.
- Writing $\int_0^1 \frac{4}{1+x^2}\,\mathrm{d}x = 4\ln 2$ (confusing it with $\int \frac{2x}{1+x^2}\,\mathrm{d}x$); it is $4\arctan 1 = \pi$.
- Proving only $\leq$ in the final step. The question asks for strict inequalities, so say why the integrals cannot be equal.

Beyond STEP

This beautiful integral, discovered in the twentieth century, shows at a stroke that the schoolbook approximation $\frac{22}{7}$ overestimates π. The same idea, integrating $x^n(1-x)^n$ against simple functions and showing the result is small but positive, drives the classical proofs that π and e are irrational: an integer between 0 and 1 cannot exist. Integrals like $\int_0^1 x^m(1-x)^n\,\mathrm{d}x$ are values of the beta function, met in first-year analysis and

probability. For the careful handling of strict inequalities between integrals, see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.